

Experiment – 2

Analysing Android App Permissions and Mobile Traffic

1. Aim

To analyse Android application network traffic and permissions by intercepting HTTP and HTTPS requests using an Android Emulator and Burp Suite.

2. Objective

- To create and run an Android Virtual Device (AVD).
- To configure a proxy between the Android Emulator and Burp Suite.
- To capture HTTP traffic from the Android device.
- To intercept and analyse HTTPS traffic by installing the Burp Suite CA certificate.

3. Required Tools / Software

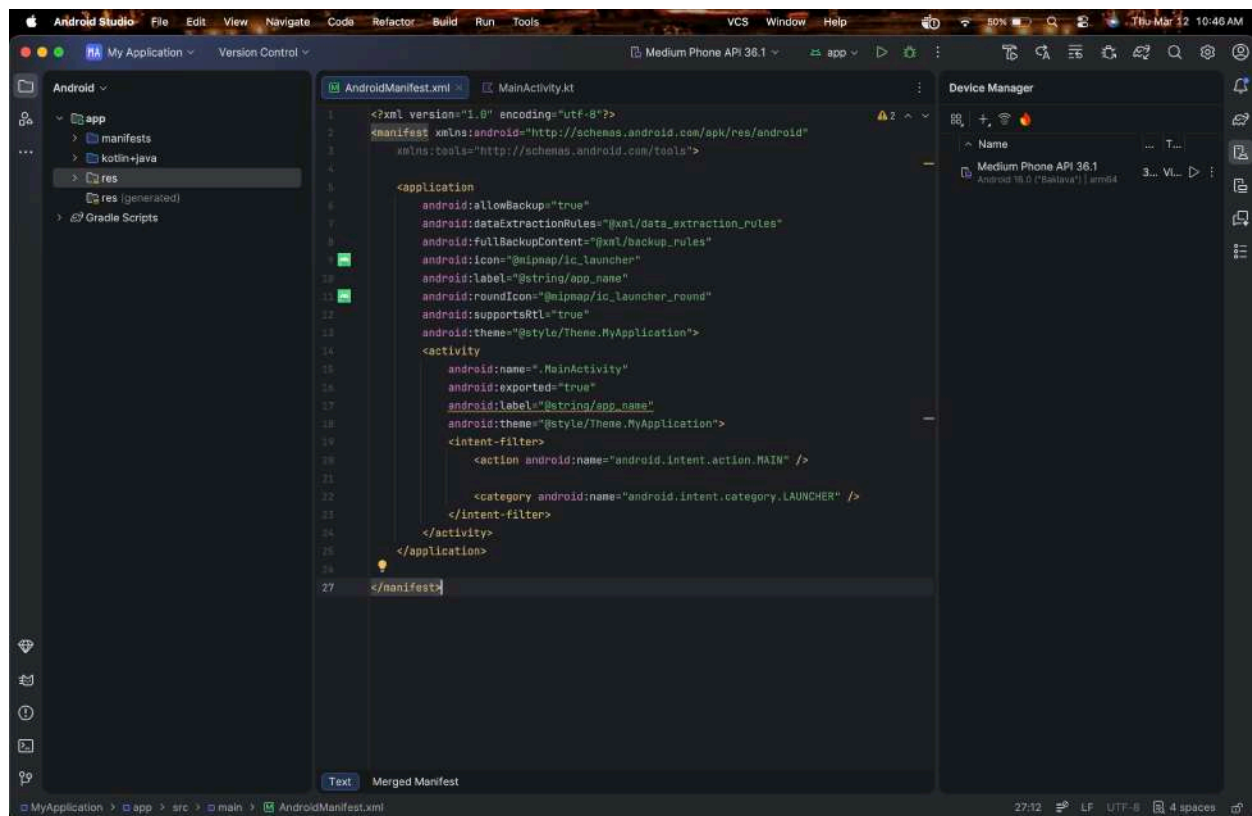
- Android Studio
- Android Emulator (AVD Manager)
- Burp Suite
- ADB (Android Debug Bridge)
- Google Chrome (inside emulator)
- Internet connection

4. Procedure

Step 1: Create Android Virtual Device

1. Install **Android Studio**.
2. Open **AVD Manager**.
3. Create a new Virtual Device.

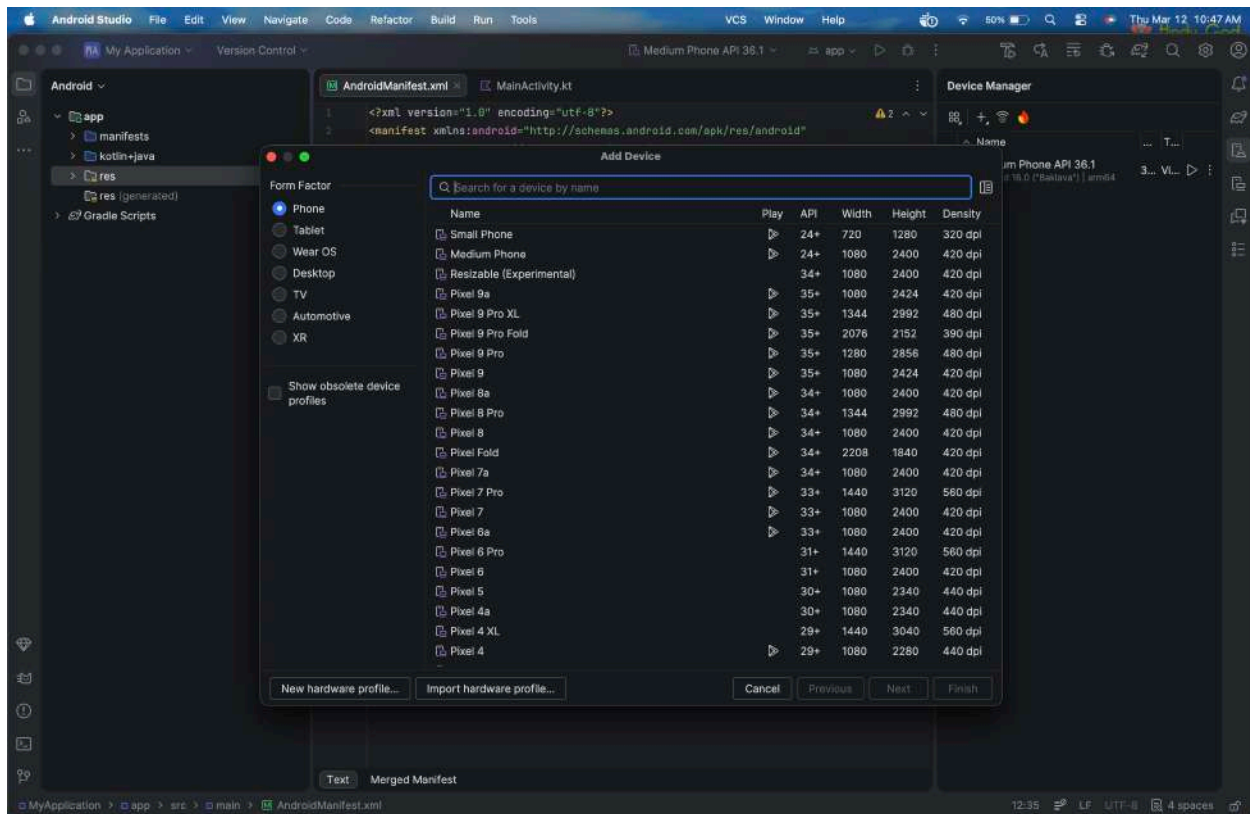
Step-1: Download Android Studio->Create Virtual Device



Step 2: Select Device

Choose any device such as:

- Pixel 2
- Pixel 9



Step-3: Start AVD from Command Prompt1

C:\Users\Admin>cd C:\Users\Admin\AppData\Local\Android\Sdk\emulator

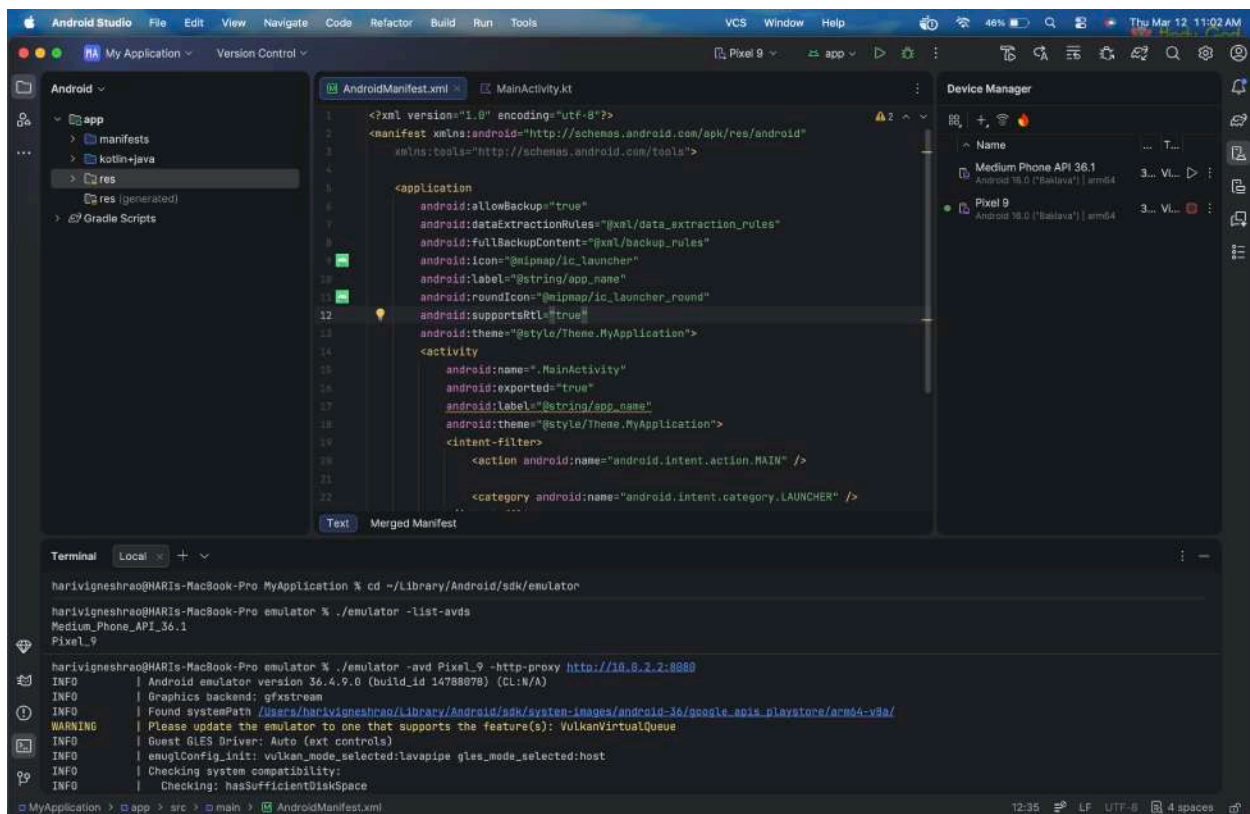
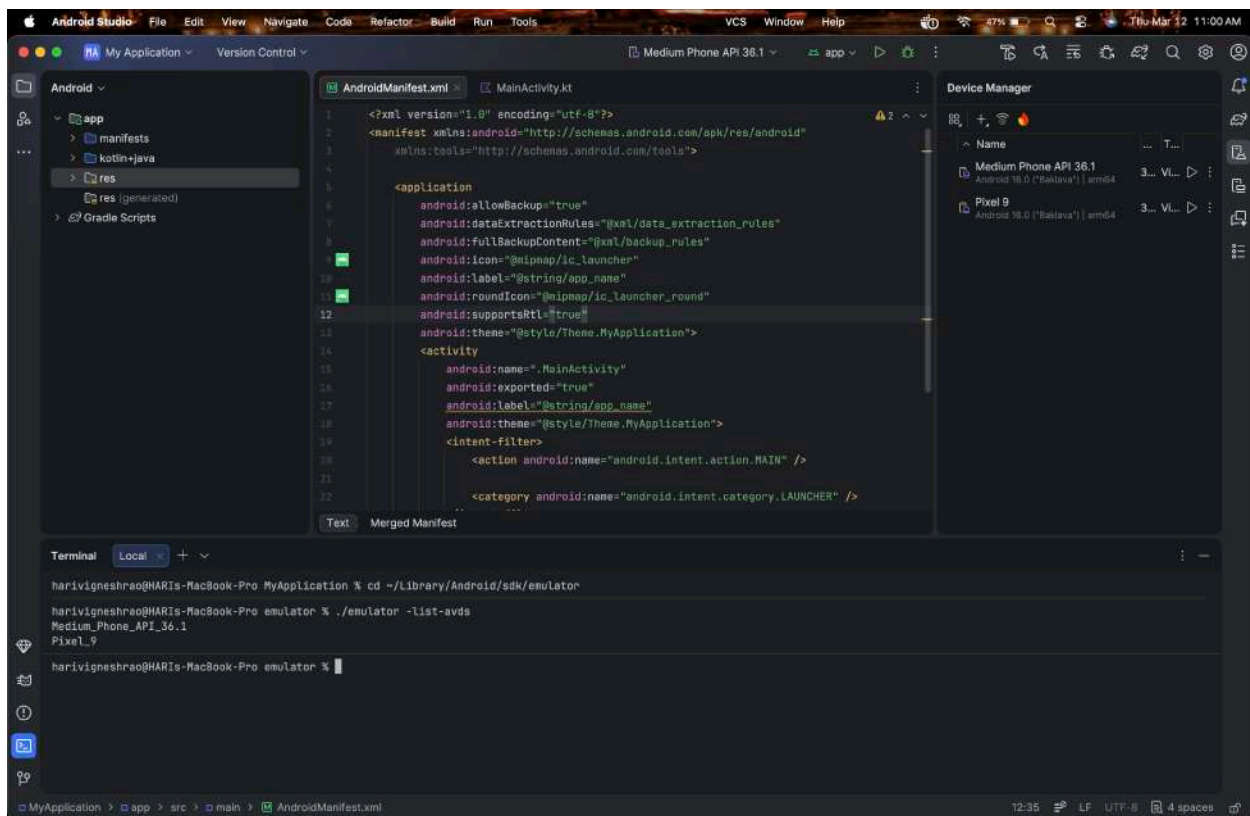
C:\Users\Admin\AppData\Local\Android\Sdk\emulator>emulator -list-avds

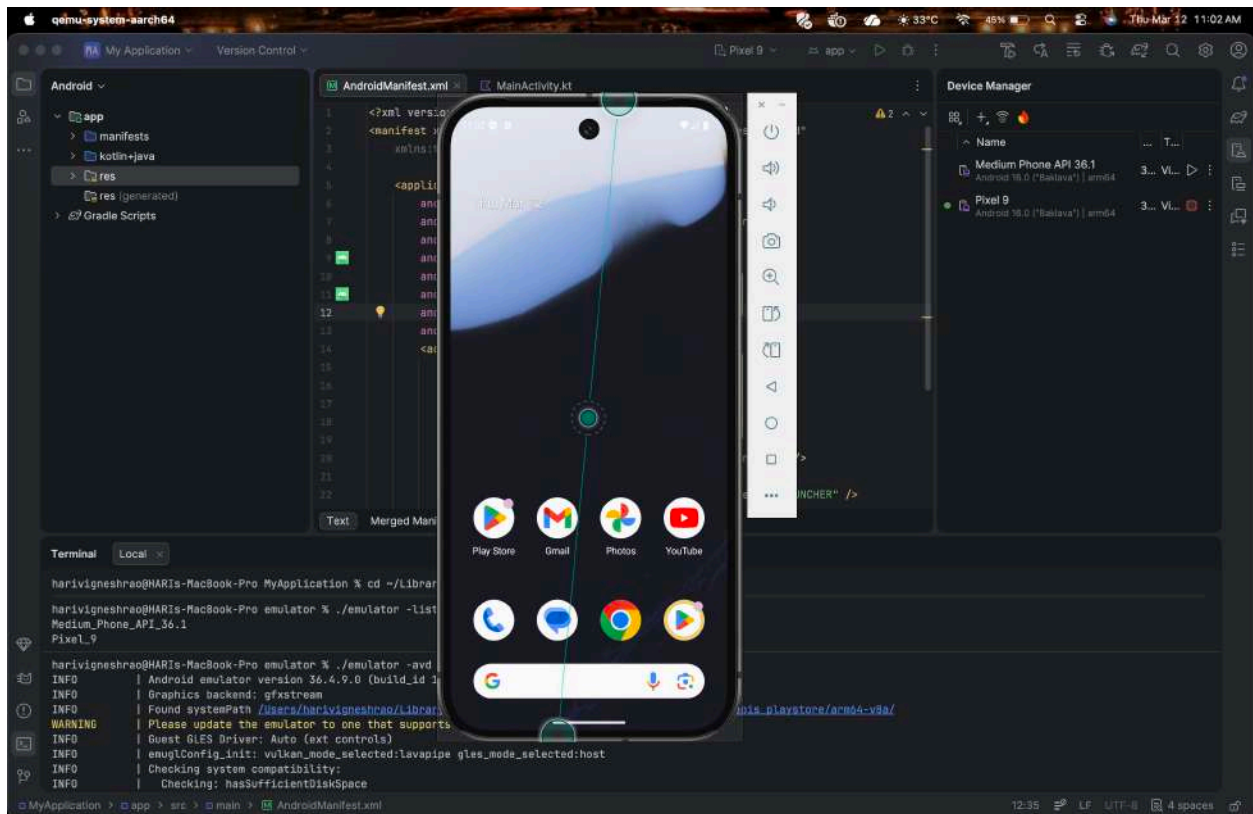
Medium_Phone_API_36.1

Pixel_2

Pixel_9

C:\Users\Admin\AppData\Local\Android\Sdk\emulator>emulator -avd Pixel_9 -http-proxy
http://10.0.2.2:8080

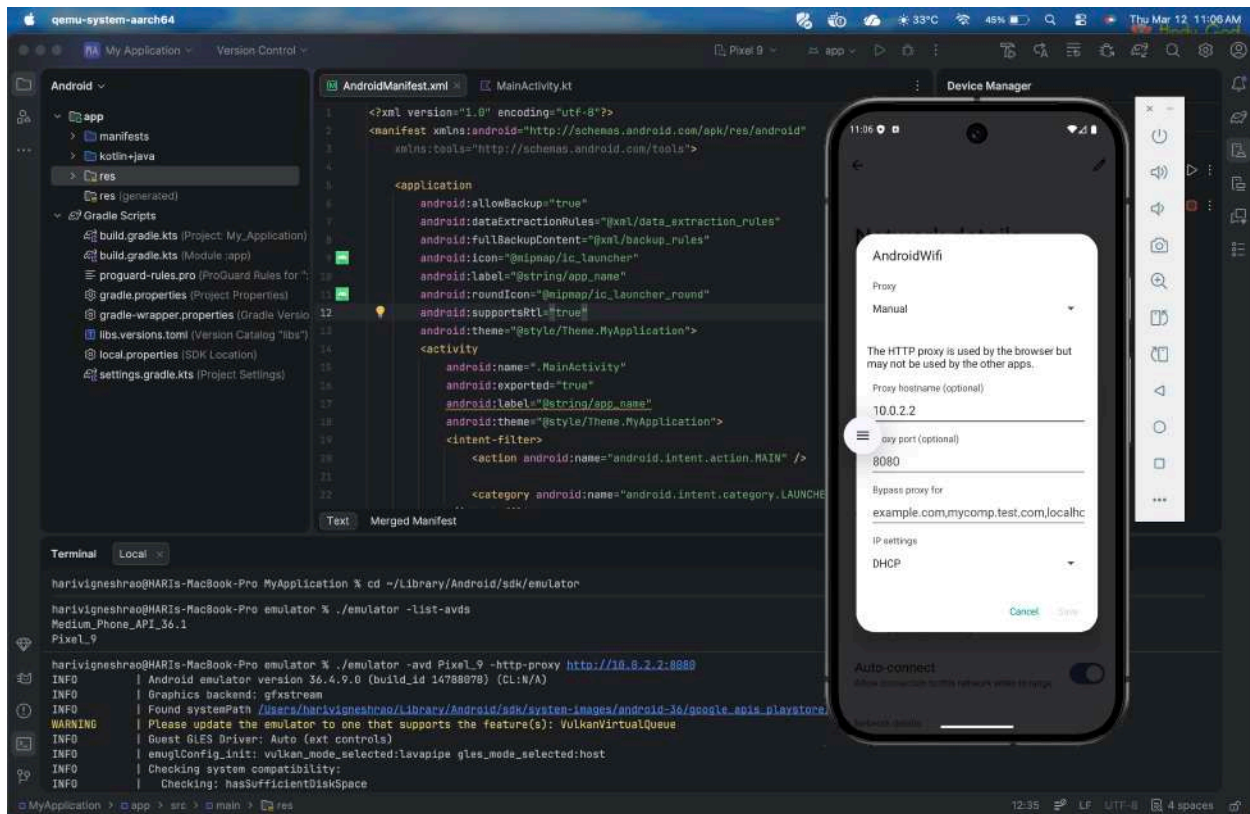




Step-4: Setup Proxy:

In AVD

Settings-> Network and Internet-> internet->select icon -> top right corner there is edit->advanced options -> proxy->Change proxy None->Manual->Hostname as 10.0.2.2 ->save



Note:if save option is not working then follow the steps in command prompt

Command Prompt2:

C:\Users\Admin>cd C:\Users\Admin\AppData\Local\Android\Sdk\platform-tools

C:\Users\Admin\AppData\Local\Android\Sdk\platform-tools>adb devices

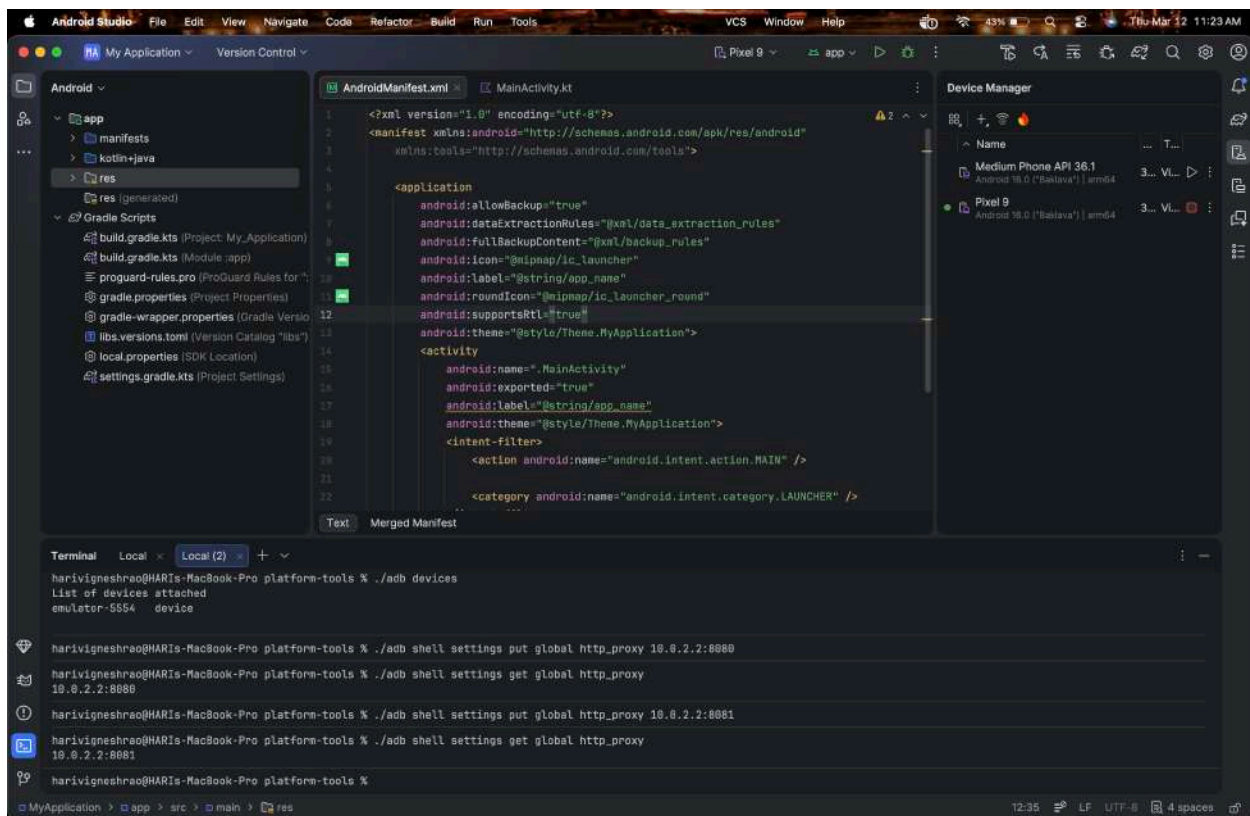
List of devices attached

emulator-5554 device

C:\Users\Admin\AppData\Local\Android\Sdk\platform-tools>adb shell settings put global http_proxy 10.0.2.2:8080

C:\Users\Admin\AppData\Local\Android\Sdk\platform-tools>adb shell settings get global http_proxy

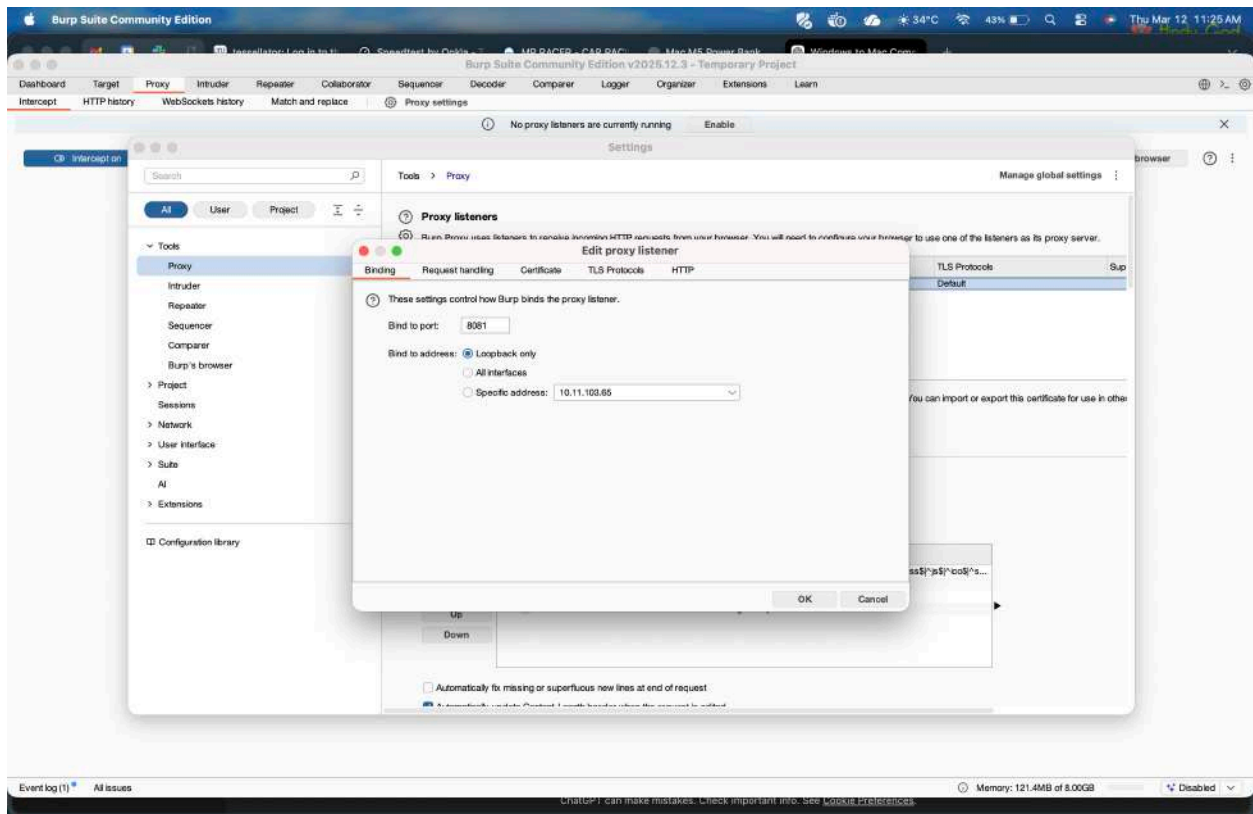
10.0.2.2:8080



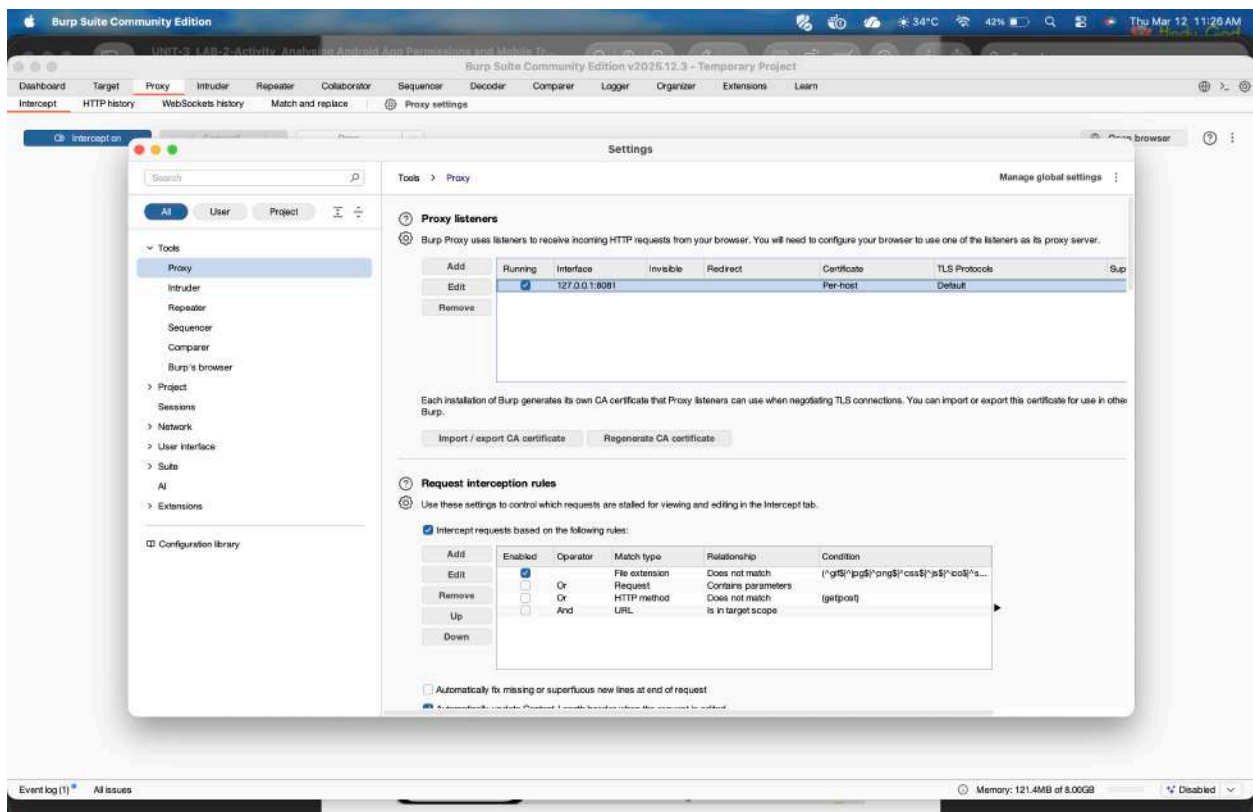
Step 5: In BurpSuit

Go proxy->options/proxy settings->under Proxy Listeners->Edit->select All interfaces->ok

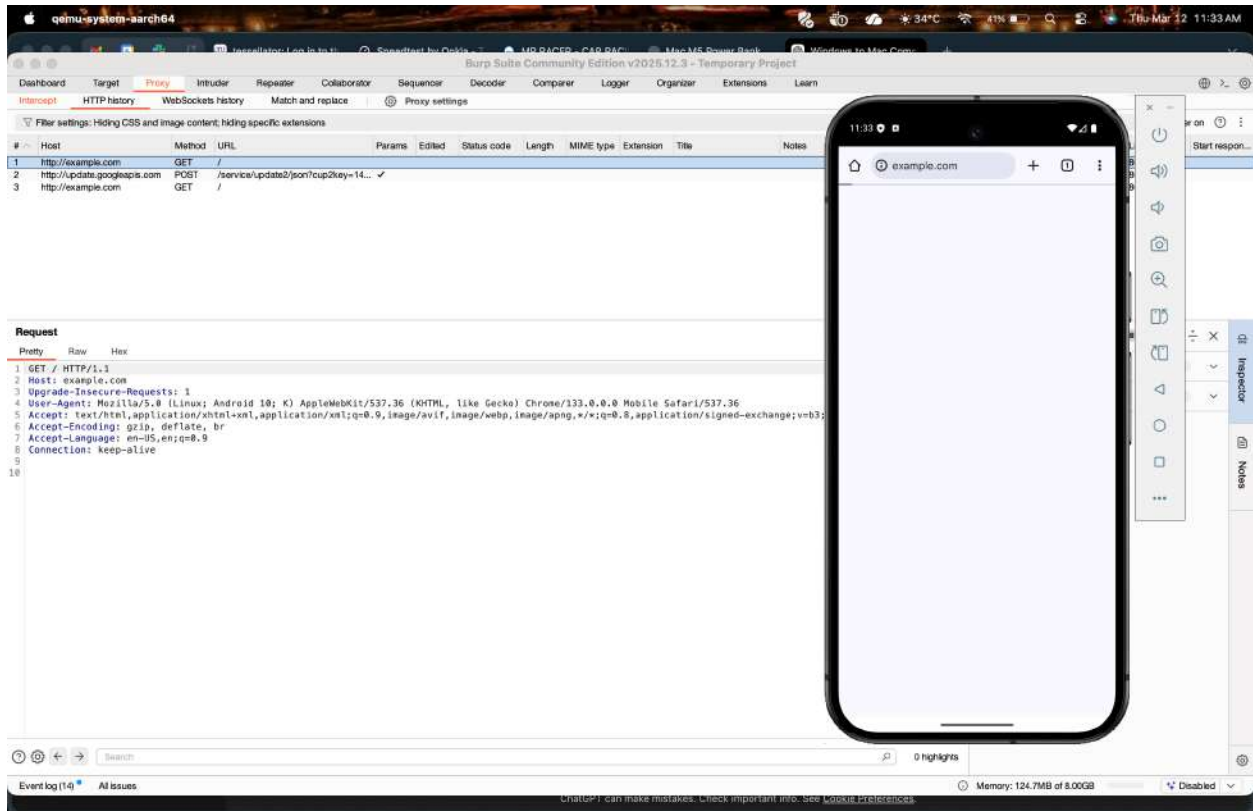
Go proxy->intercept on



You can see BurpSuite is accepting all interfaces traffic

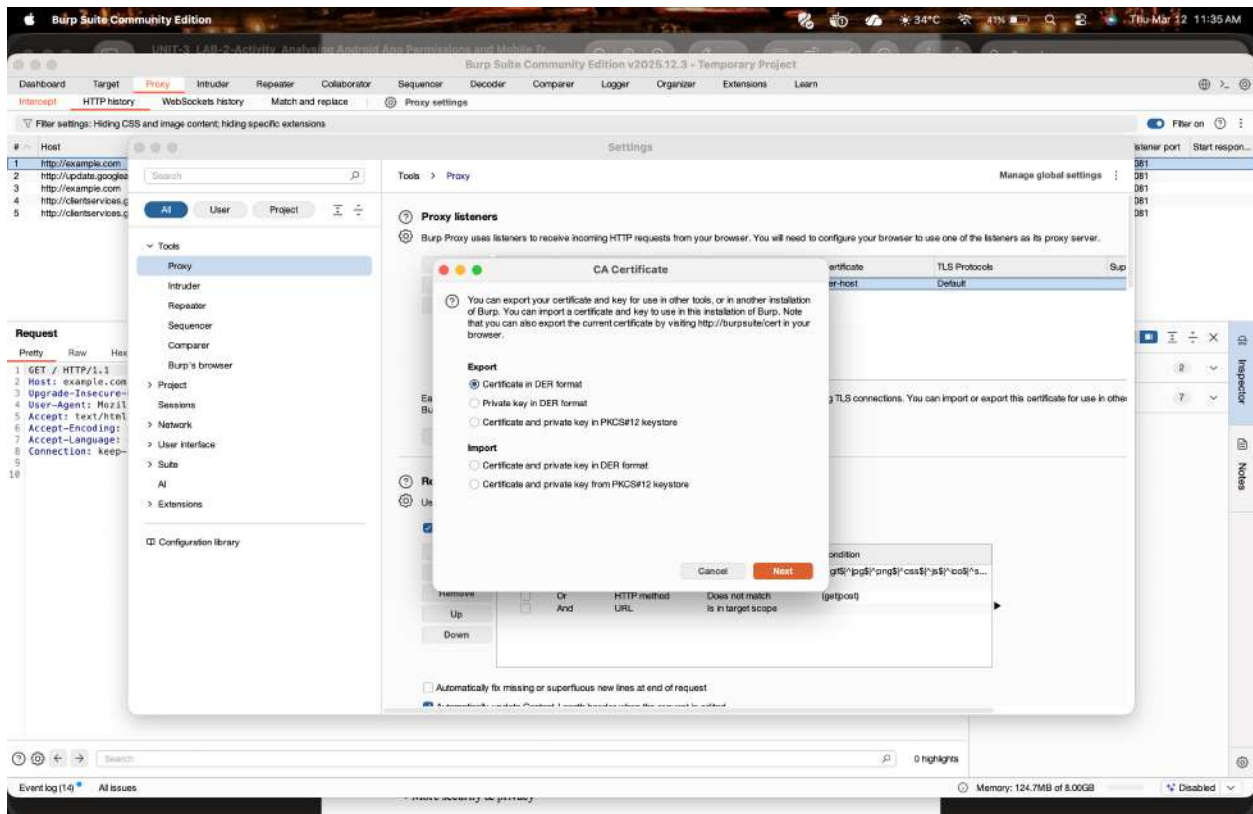


Step 6: in AVD type in chrome example.com and you can see burpsuit HTTP history



Step 7: in Step 5 we only able to http request and response only but if you want to analyse https request and response follow the Procedure to download CA certificate from BunrpSuit.

Go proxy->options/proxy settings->import/export CA certificate->Certificate in DER format->select file-> Downloads->give name as burpcer.der

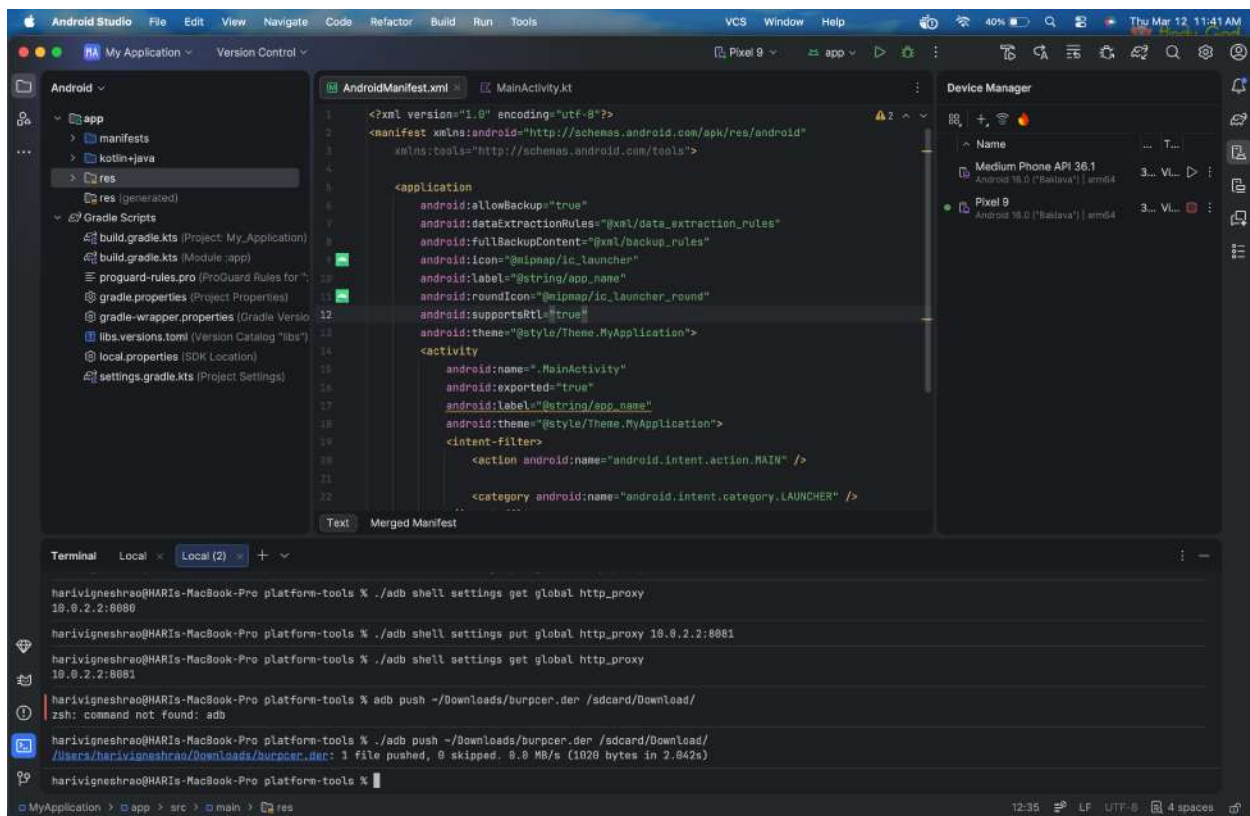


Step 8: Sending CA certificate to AVD

Go to command Prompt

```
C:\Users\Admin\AppData\Local\Android\Sdk\platform-tools>adb push
```

```
C:\Users\Admin\Downloads\burpcer.cer /sdcard/Download/
```

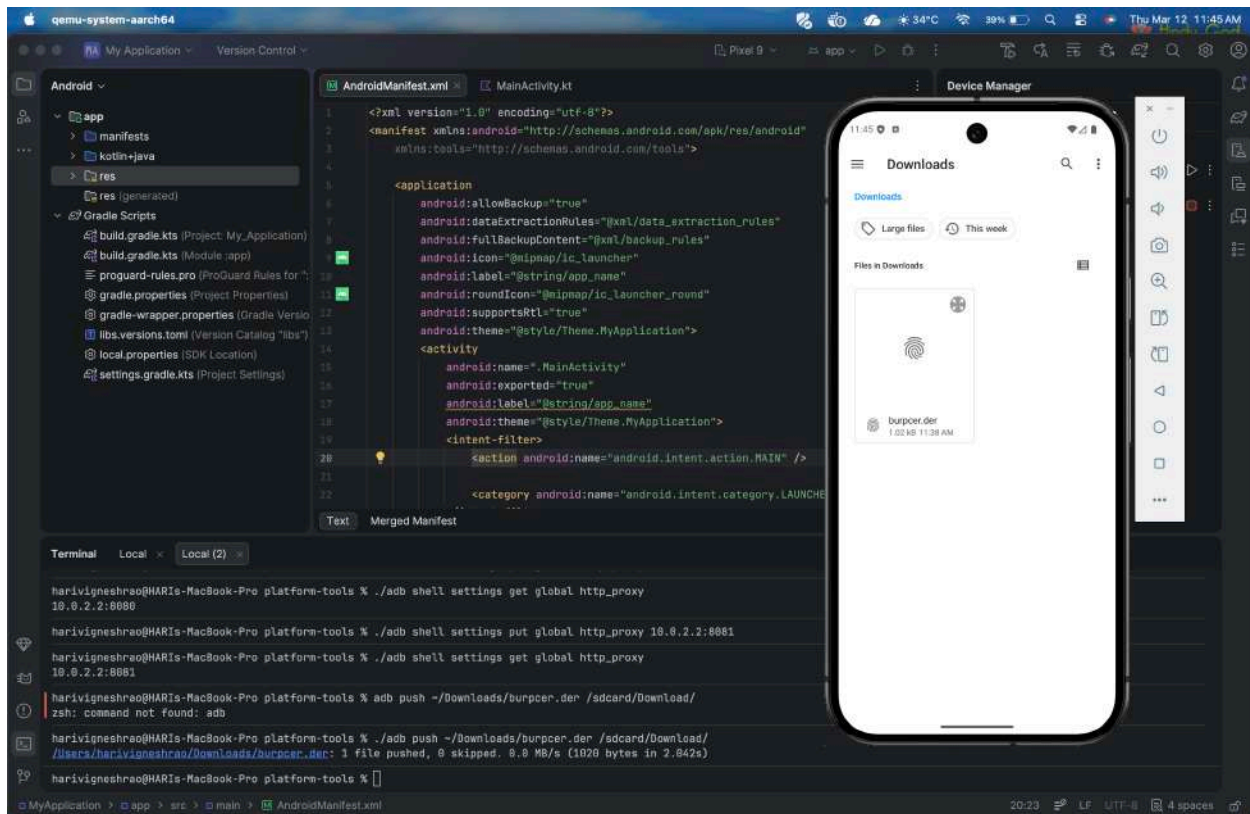


To install Certificate in AVD

Settings

- Security & privacy
- More security & privacy
- Encryption & credentials
- Install a certificate
- CA certificate

In AVD we can see burpcer.der certificate

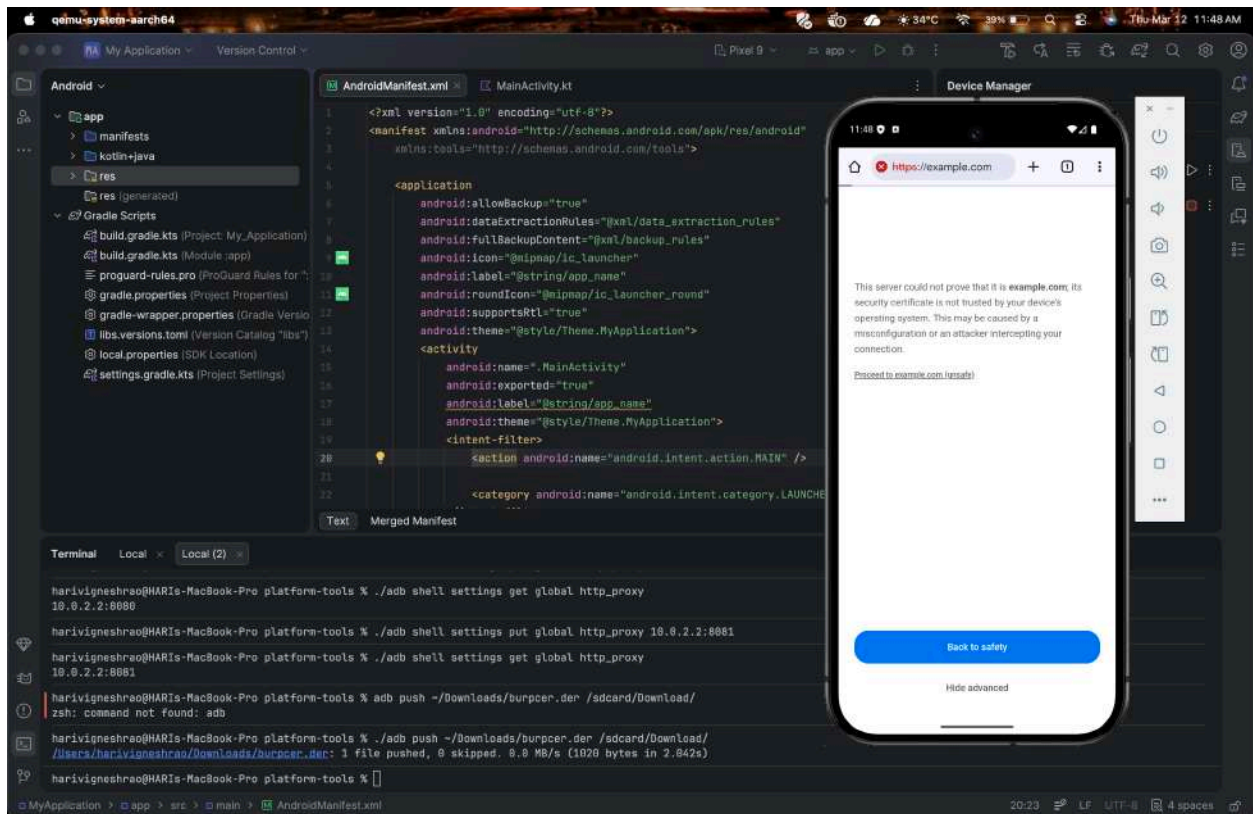


Step 9:verify HTTPS Interception

1. Open browser in emulator.
2. Visit any HTTPS website.

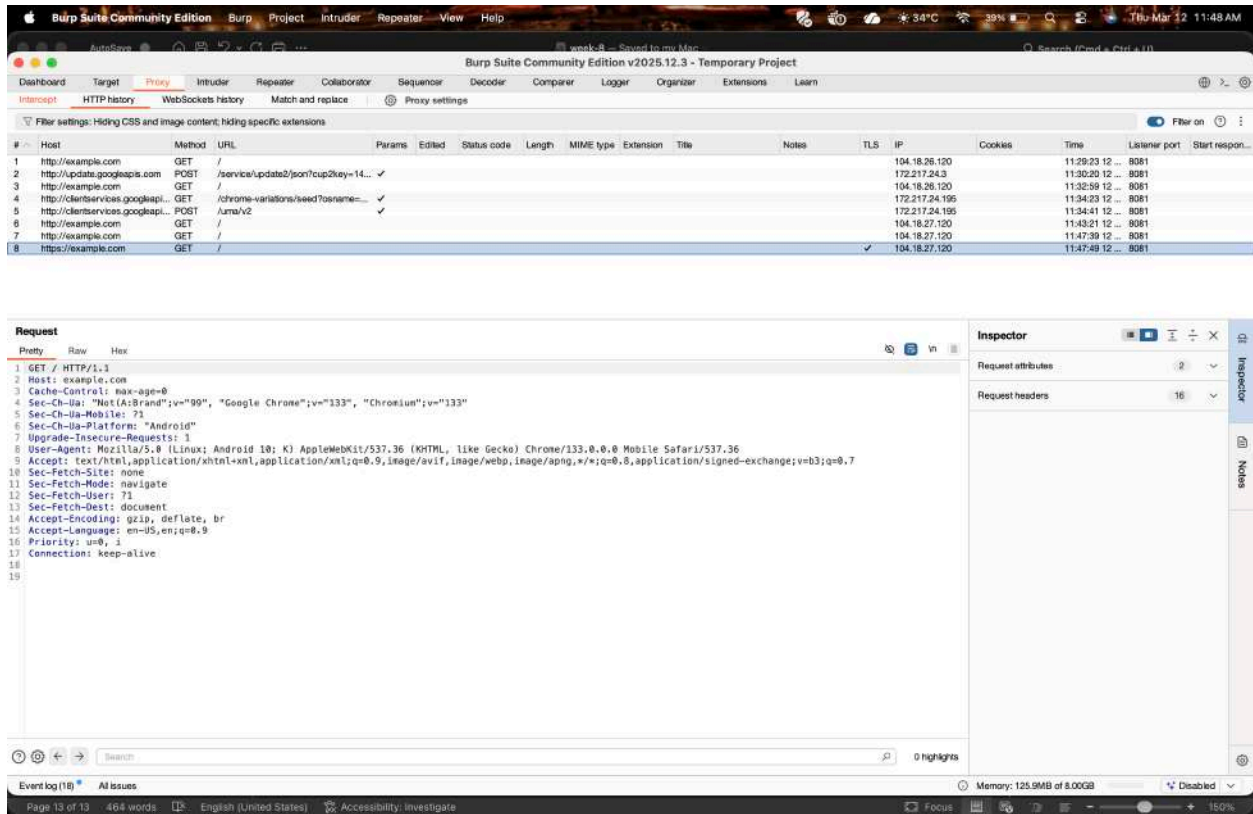
Example:

<https://example.com> or <https://wikipedia.com>



3. In Burp Suite, check **Proxy** → **HTTP History**.

Now HTTPS requests and responses will be visible.



5. Result

Android application network traffic was successfully intercepted and analysed using Burp Suite by configuring proxy settings in the Android emulator and installing the Burp Suite CA certificate.

6. Precautions

- Ensure Burp Suite proxy port (8080) is correctly configured.
- Verify the emulator is running before executing ADB commands.
- Install the CA certificate properly to capture HTTPS traffic.
- Disable intercept when not required to allow traffic to pass normally.